

Matt Lasselle

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Education:

Iowa State University, College of Engineering

Bachelor of Science, Mechanical Engineering | Dean's List Spring 2023, 2024, 2025

Expected Graduation: May 2026

GPA: 3.42

Professional Experience:

Stratasys – Test Engineer Intern

May 2025 – August 2025

- Developed and programmed a compression test method on a MTS Criterion Model 43 using TestSuite TW Elite, enabling improved data logging and more accurate measurement of a wide range of material properties.
- Assessed material data of 3D printed polymers to establish new metrics for defining compressive yield strength to improve data provided to customers.
- Reduced required compression testing specimens by analyzing the effect of slenderness ratio on compressive yield strength, while maintaining ASTM D695-10 compliance.
- Automated data formatting to increase post-test analysis efficiency and implemented safety measures in test methods to improve lab safety and protect equipment and operators.

Technical Projects:

IREC 2026 Competition Rocket

June 2025 – Present

- Led the design and simulation of the competition flight vehicle consisting of over 500 independent components
- Performed load frame testing on composite and additively manufactured components to evaluate tensile, compressive, bending, high-temperature, and creep failure behavior.
- Fabricated subsystem components from aluminum, stainless steel, composites, and plastics using machining, additive manufacturing, and molding processes
- Developed two test vehicles to serve as primary platforms for validating and iterating competition rocket subsystems

Dual Extrusion 3D Printer

August 2025 – Present

- Designed and manufactured a modular dual-extrusion 3D printer that uses a servo-actuated secondary nozzle
- Performed thermal analysis on the toolhead assembly incorporating copper, stainless steel, and ABS components.
- Validated interlayer bonding strength between materials printed from each nozzle through tensile testing.

Solid Rocket Engine and Test Stand

November 2025 – January 2026

- Modeled a solid rocket motor and ground-test stand to measure thrust, chamber pressure, and casing temperature during hot-fire testing.
- Validated combustion chamber and seals through hydrostatic testing up to 1.5 times the expected pressure.
- Developed an additively manufactured nozzle incorporating a graphite throat to mitigate erosion.

Mechanical Load Frame

May 2025 – November 2025

- Designed and built a mechanical load frame capable of recording load, displacement, and time data, with a peak capacity of 1,200 lbf and 6.5" of travel.
- Wired and programmed the electrical system, including stepper motors, motor drivers, OLED displays, load cells, control buttons, and Arduino microcontrollers

Piston Ejection System

June 2024 – June 2025

- Designed a 2" diameter piston with focus on failure modes produced from over-pressurization of the piston cylinder.
- Validated piston system through hydrostatic testing to a proof of 2 times the expected operating pressure
- Fabricated piston components from aluminum utilizing the mills and lathes

Leadership:

Vice President of Engineering of Cyclone Rocketry Club

June 2025 – Present

- Oversaw all engineering activities for the club, reviewed and approved critical design decisions, and led the development of subsystem technical documentation and reports
- Served as the flyer of record, ensured full competition compliance, and led system-level testing and validation

Mechanical Team Lead of Cyclone Rocketry Club

June 2024 – June 2025

- Led a team of 5 students in designing and testing a piston ejection system to separate a 70 lb rocket and deploy the main parachute.

Technical Skills and Certifications:

Mechanical Design	Test Method Development	Testing and Instrumentation	Data Analysis	SolidWorks	ANSYS Mechanical
Additive Manufacturing	Design for Manufacturing	Fusion 360 CAM	DC Electronics	Excel	Tripoli L3 Certified